Khaled Lakhdher

Computer Engineer | AI & Data Science | Full-Stack AI Developer

Sousse, Tunisia | ing.khaledlakhdher@gmail.com | +216 29 843 899 | in/khaled-lakhdher

khaled-lakhdher-portfolio.vercel.app | github.com/KhaledLakhdher

Profile

I'm an AI and data engineer who turns models and data into dependable products. I build LLM and RAG systems, multi-agent platforms, and end-to-end ML pipelines, with a metrics-first focus on data quality, evaluation, and clean deployment. Comfortable across the stack: Python, SQL, FastAPI, and Docker.

Experience

AI Engineer August 2026 - Present · Hammamet

CTT - Carthage Training & Technologies

- Continued on **Hotel Mind** full-time after the end-of-studies internship, shipping new features and migrating the platform's data layer onto Nextcloud.

Artificial Intelligence Developer Intern (End-of-Studies Project) February 2026 - July 2026 · Hammamet

CTT - Carthage Training & Technologies

- Designed and developed **Hotel Mind**, a multilingual, multi-agent hotel search and booking assistant: 9 LangChain/LangGraph agents call 32+ tools across 10 real SOAP endpoints for live hotel, room, and pricing data.
- Grounded every answer in live tool output, not model memory, via a hybrid flow (AI search, a guest-detail form, a sandboxed SOAP reservation) on a full-stack FastAPI + Next.js app with LangSmith tracing.

Artificial Intelligence Developer Intern July 2025 - September 2025 · Hammamet

CTT - Carthage Training & Technologies

- Fine-tuned a Hugging Face NLP model to classify customer reviews into three sentiment classes, improving accuracy by 25% over the baseline.
- Built a Django backend to manage customer data and automate export of analysis results to Excel for reporting.
- Owned the end-to-end Python pipeline for preprocessing, training, and integration.

Projects

Nexus AI - Artificial Intelligence Employee Management Platform

- Built a multi-tenant platform where LLM agents are governed like employees: scoped IAM-style permissions, hierarchical vector memory, multi-agent orchestration with real-time WebSocket monitoring, and a provider-agnostic model gateway (Anthropic, OpenRouter, Gemini) running agents for a fraction of a cent per run. Stack: FastAPI, PostgreSQL/pgvector, Redis, Next.js.

DecisionFlow - AI Business Analyst (Multi-Tenant SaaS)

- Turned an uploaded CSV into a data warehouse: automated cleaning, a star schema, forecasting, and natural-language Q&A, built on a multi-tenant SaaS (FastAPI, PostgreSQL/RLS, DuckDB, Polars, Next.js) with LLM-generated SQL sandboxed across 3 layers and tenant isolation enforced in-database.

CodeReview Evaluation - Benchmarked LLM Code Reviewer

- Developed an LLM code-review and evaluation framework on BugsInPy (501 real bugs); on a 49-bug run, an LLM-as-judge showed a single detection score overstates capability by ∼1.8× (78% localization vs. 43% diagnosis), compared across 3 model tiers, with a live Streamlit demo.

Certifications

- Building Transformer-Based Natural Language Processing Applications - NVIDIA (Dec 19, 2025)
- Fundamentals of Deep Learning - NVIDIA (Nov 22, 2025)
- Applications of AI for Predictive Maintenance - NVIDIA (Oct 27, 2025)

Skills

Programming Languages: Python · SQL · TypeScript · JavaScript · Java · C

AI & LLM Engineering: PyTorch · TensorFlow · Transformers · Hugging Face · scikit-learn · LangChain · LangGraph

Backend & Data: FastAPI · Django · Flask · PostgreSQL · Pandas · NumPy

Frontend, Cloud & Tools: Next.js · React.js · Tailwind CSS · Docker · AWS · Git · GitHub · Power BI · Streamlit · Postman

Concepts & Methods: Large Language Models (LLMs) · RAG (Retrieval-Augmented Generation) · AI Agents · Agentic AI · Multi-Agent Systems · Natural Language Processing (NLP) · Machine Learning (ML) · Deep Learning (DL) · Agile/Scrum

Education

Computer Engineering - AI & Data Science Specialization 2023 - 2026

EPI - International Polytechnic School · Sousse, Tunisia

Preparatory Cycle 2021 - 2023

EPI - Private International Polytechnic School · Sousse, Tunisia

Languages

French: DELF B2

English: TOEIC 825/990

Arabic: Native